Prior Authorization Decision Support: AI Product PRD

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Scope note: the working prototype (React + Vite, scripted data, six cases, a 106-page synthetic packet, and a patient portal) is the MVP described here. Production-scale items are labeled as target state. All figures are targets or hypotheses to validate, not measured results.

What changed in v1.1 (marked NEW below)

- Source packet expanded from 14 to 106 pages, closer to a real faxed submission: 24 physical therapy visit notes, imaging, injection records, consents, a blank fax page and a resent page. Citations come from a generated page index and are checked by script.
- Packet viewer rebuilt for long packets: cited page renders first, the rest load as you scroll, with page jump and a section index.
- New user: the patient. A patient portal shows status, the decision clock, what the plan is waiting on, costs, payment choices, preparation steps, and appeal rights after a denial.
- Patient actions are written to the case audit trail, and a live Patient Engagement report is added.
- Landing page rewritten: the end-to-end journey (doctor's office → AI → UM nurse → medical director → decision → patient) at the top, and two entry points, UM nurse view and Patient view.

1. Problem (AI-Specific)

- **Problem:** Utilization management (UM) nurses decide whether a requested inpatient admission meets level-of-care criteria. The evidence sits in faxed packets that often run past 100 pages (cover sheet, primary care and specialist notes, imaging, therapy records, H&P, cardiology clearance, labs, surgeon orders). Finding each fact, checking it against layered coverage policy, and documenting the rationale is slow, and decisions run against regulatory clocks (CMS: 7 days standard, 72 hours expedited).
- **Patient side:** NEW While the request is pending, the patient usually can't see where it stands, who owes the next step, what they will owe, or what to do after a denial. That gap becomes phone calls to the plan and the provider office, and surprise bills. CMS-0057-F requires payers to give a specific reason for every denied prior authorization (from 2026) and to add prior authorization information to the Patient Access API (by January 1, 2027).
- **Why AI:** Inputs are unstructured and inconsistent across providers, and the same fact appears in different words ("multi-day stay" vs. "2-midnight admission"). Rules alone cannot read the packets. A model can extract and map facts, but only if every claim is traceable and uncertain

cases reach a human. At 100+ pages, the key sentence can sit on page 95, so finding it matters as much as reading it.

- **Status quo:** Manual read-through and copy-paste into a checklist. Inconsistent depth of review, no uniform citation trail, and clarification requests to providers are written ad hoc. Patients get updates by letter or phone.

2. User

- **Primary persona:** UM nurse reviewer. **Secondary:** medical director (handles escalated and denial decisions). **Tertiary:** auditor or compliance reviewer reading the trail afterward. **Fourth:** NEW the patient (plan member) waiting on the decision, often with a family caregiver helping.
- **Job to be done (reviewer):** Reach a defensible determination (approve, request clarification, escalate) quickly, with evidence to back it.
- **Job to be done (patient):** NEW Know where my request stands and what happens next; get ready for the procedure; understand and settle what I'll owe; and if it's denied, know why and how to appeal.
- **AI interaction pattern:** Suggestion-and-approve. The AI extracts, maps, and drafts; a human makes every determination. No auto-approval or auto-denial. For the patient, a status view plus self-service actions; the help assistant answers only status, cost, and preparation questions.
- **Trust requirements:** High. Every checkmark must be verifiable in one click against the source page and sentence. Denials are made only by the medical director. The patient never sees internal AI summaries or reviewer comments, only plain-language status and reasons.

3. Core Metric

- **North Star:** Time from packet receipt to a defensible determination, with zero uncited findings.
- **Baseline:** Not yet measured. First task in the pilot is to time the current manual process on a sample of cases.
- **Target (hypothesis):** At least 40% reduction in nurse review time on straight-through cases, with 100% of findings carrying a source citation.
- **Patient metric (hypothesis):** NEW At least 70% of approved patients confirm their cost estimate before the procedure date, and 30% fewer inbound "where is my authorization" calls.
- **Leading indicators:** Citation click-through rate; share of cases routed to each review tier; clarification round-trip count; reviewer override rate on AI criterion status; patient portal views, estimate acceptance, pay now vs. later, declines by reason, callback volume, and appeal requests.

4. MVP Features

Feature	Priority	Description

Packet extraction	P0	Pull procedure, comorbidities, clearance, and expected length of stay from the packet, each tagged with document and page
Hierarchical criteria checklist	P0	Four inpatient level-of-care criteria, each marked Met / Ambiguous / Missing / Not Met, mapped to the governing tier (NCD, LCD, payer policy, MCG)
Verifiable citations	P0	Clicking a citation opens the real packet PDF at the page with the cited sentence highlighted; public CMS policy links out to the CMS Coverage Database
Realistic long packet NEW	P0	106-page synthetic fax generated by script, with key evidence deep in the stack (length of stay on p. 95) plus real-world noise: a blank page and a resent page. A generated index holds every citation's page and quote, and a check script confirms each quote is on its page
Risk-scoped review tiers	P0	Banner explaining in plain language: lighter-touch review, standard review (one gap), or escalation (two or more gaps, or any Not Met)
Determination actions	P0	Approve, request provider clarification, escalate to medical director, deny (medical director only), each with an AI-drafted, human-readable summary shown before confirming
Audit trail	P0	Timestamped record of every action and the exact AI summary shown, exportable. Now also records the patient's portal actions
Landing page and entry points NEW	P1	Explains the product end to end in six steps, each labeled with who acts (doctor's office, AI, UM nurse, medical director, health plan, patient), then offers two entry points: UM nurse view (case queue) and Patient view (patient picker). A "Why it matters" section below compares today's manual process with the future state
Patient status and decision clock NEW	P0	Five-step tracker (initiated by doctor's office, sent to insurance, review in progress, additional information needed, approved), live from the same case record, with the decision timeline: critical 72 hours or standard 7 days, and time remaining
What we're waiting on NEW	P0	When information is needed: who owns it (the doctor's office, not the patient), what was asked in plain language, and whether it was delivered and seen
Denial reason and appeal path NEW	P0	The specific unmet criteria in plain language, appeal rights and the 65-day deadline, an "ask my doctor to appeal" request, and doctor-to-doctor (peer-to-peer) review status

Cost estimate and payment choice NEW	P0	After approval: copay, deductible and coinsurance with plain-language definitions and the out-of-pocket maximum. Accept, then pay now (card or HSA/FSA) or later (at the appointment, or a 3, 6 or 12-month interest-free plan). Decline routes to the care team; if cost is the reason, a financial-assistance check runs first
Ask about this document	P1	In-page chat over the packet with citation links back to source, extended for imaging, therapy and injection history
Packet viewer at scale NEW	P1	Renders the cited page first and loads the rest as you scroll, so a 100+ page packet opens instantly; page jump and section index
Parse-failure path	P1	Illegible packets are not guessed at; the case routes to manual review with a provider resubmission request. The patient sees that the doctor's office must resend pages
Case queue and CMS clocks	P1	Queue by status (needs action, escalated, closed) with standard and expedited deadline indicators
Get ready NEW	P1	After approval: an interactive pre-op checklist taken from the surgeon's orders (nothing to eat after midnight, which medications to hold), a countdown, appointment details, and an add-to-calendar file
Patient activity and documents sent NEW	P1	Plain-language history of every step, and the document types and page counts shared with the plan
Patient help assistant NEW	P1	Answers status, cost, payment, preparation and appeal questions from the live case. Medical questions go to the care team, never answered
Language and accessibility NEW	P1	English and Spanish, read-aloud, mobile-first layout
Callback and notifications NEW	P2	Request a call from the doctor's office or the plan; choose text or email updates, with a notification inbox (simulated delivery)
Reports view	P2	Aggregate view of cases by tier and outcome, plus a live Patient Engagement tab: acceptance rate, pay now vs. later, declines by reason, callbacks, appeals

5. Constraints

- **Technical:** Prototype is front-end only with no backend. Production needs OCR for faxed scans, PDF text-layer mapping for highlighting, and page-level latency suitable for interactive use. At 100+ pages, extraction must work page by page with retrieval, not one pass over the whole document.

- **Data:** Prototype uses synthetic, scripted cases only. Real use requires PHI handling and policy content licensing (MCG and InterQual are proprietary). Real cost estimates need the plan's benefit and accumulator data (deductible and out-of-pocket met to date) and the facility's price.
- **Regulatory:** HIPAA; CMS decision timeframes; payer and state UM rules. The tool supports clinical judgment and is not a determination engine. The cited LCD in the demo is illustrative. NEW CMS-0057-F: specific denial reasons and prior authorization data in the Patient Access API. Medicare Advantage appeals: members have 65 days to request reconsideration; plans decide standard pre-service appeals within 30 days and expedited within 72 hours, and must expedite when a physician asks. Payments fall under PCI DSS; the demo collects no card data. Language access rules apply to member communications.
- **Organizational:** Single builder, portfolio-scale scope. No claim of clinical validation.

6. Grounding Strategy

- **Source of truth:** The submitted packet for patient facts; a versioned criteria library for policy (NCD, then LCD, then payer policy, then MCG, in precedence order).
- **Retrieval approach (target):** Classify pages first, then run extraction only on relevant sections, constrained to structured fields that each require a page and quoted span; criteria evaluation against a fixed, versioned rule set rather than free-form recall of policy. Prototype scripts this behavior.
- **Freshness:** Criteria library versioned with effective dates; each determination records the version used.
- **Citation and traceability:** Mandatory. A finding with no verifiable quote cannot be marked Met. In the prototype, every citation's page comes from the packet index, never a hard-coded number.
- **Patient view:** NEW Derived entirely from the same case record the nurse acts on, with no separate copy of status. Each unmet criterion maps to one plain-language denial reason.

7. Hallucination Guardrails

- **Detection:** Automated check that every quoted span exists verbatim on the cited page; any finding failing this is downgraded to Missing. The prototype runs this check over the whole packet index (19 of 19 citations verified).
- **Prevention:** Structured output with required quote and source fields; the model is instructed that "not found" is a valid answer.
- **Fallback:** Ambiguous criteria are never resolved by the AI. They generate a targeted provider question (for example, "confirm expected length of stay in midnights"). Conflicting documents are flagged as conflicts, not reconciled. Unreadable packets route to manual review with no extraction attempted.
- **Patient assistant:** NEW Scripted and state-aware in the MVP. Symptom, dose or safety questions are routed to the care team, with a callback option and an emergency line, and are never answered. Internal AI summaries and reviewer comments never reach the patient.

- **User-facing signals:** Four-state status per criterion, review-tier banner, and a visible reason for each non-Met status.

8. Cost Budget

- **Cost per case (target):** Under $1.00 in inference for a 100+ page packet (revised from $0.50 at 14 pages), versus nurse time measured in tens of minutes.
- **Model tier:** Small model to classify pages; mid-tier model for extraction and criteria mapping on the relevant pages only; escalate to a larger model only for conflicting or low-confidence documents. A model-backed patient assistant, if added, uses a small model limited to the case record.
- **Volume assumptions:** To be sized with the pilot site; assume low thousands of cases per month for budgeting.
- **Cost controls:** Cache extraction per packet version, classify pages before extraction, cap pages per call, route simple cases to the cheaper tier, and re-run only changed sections after a provider response.

9. Eval Strategy

- **Offline:** Golden set of packets with labeled facts and expected criterion status. The prototype's six cases are the seed set: clean approval, ambiguous length of stay, multi-ambiguous with conflicting notes, corrupted packet, resolved-after-clarification, and expedited clarification. Expand with real de-identified packets, including long ones with duplicates and blank pages.
- **Patient-facing:** NEW Readability of patient copy (target 6th to 8th grade); native-speaker review of every translation; guardrail tests in which every medical question is routed and none is answered.
- **Online:** Shadow mode first (AI output not shown to the nurse), then assisted mode with override tracking and sampled second-reviewer audit.
- **Metrics:** Extraction accuracy and recall by page depth, citation validity (quote exists on cited page), criterion-status agreement with expert labels, false-Met rate (the critical error), latency, and patient-assistant routing accuracy.
- **Cadence:** Every prompt, model, or criteria-library change; monthly drift check on production samples.

10. Production Readiness

- **Monitoring:** Citation-validity rate, override rate, parse-failure rate, tier distribution, cost per case, and time-to-determination against CMS clocks; for the portal, notification delivery, estimate acceptance, and callback response time.
- **Rollback:** Prompt, model, and criteria-library versions pinned per release; one-step revert to the prior version.

- **Abuse and cost prevention:** Authenticated roles (nurse, medical director, auditor, member), identity proofing for member access, per-user rate limits, and page-count caps. Payments through a PCI-compliant hosted payment page.
- **Launch checklist:** False-Met rate below the agreed threshold on the golden set; security and privacy review; medical director sign-off on the escalation rules; on-call coverage for clock-critical cases; legal review of denial and appeal wording; accessibility review (WCAG 2.2 AA); native-speaker sign-off on Spanish.

11. Engineering Context

- **Architecture:** Browser app with case queue, case workflow (extraction, criteria, review, determination), packet viewer, audit trail, and a patient portal reading the same case record. Target adds an extraction service (OCR, page classification, structured extraction), a criteria service (versioned rules), a secured audit store, and a notification service.
- **Integrations (target):** Fax or portal intake, EHR and payer systems of record, identity provider, Patient Access API (FHIR), benefits accumulators and facility pricing for estimates, SMS and email delivery, and a payment processor.
- **Team and ownership:** PM and builder (this project); clinical SME and compliance reviewer needed for a pilot; member-experience and billing owners for the portal.
- **Dependencies:** De-identified real packets, licensed criteria content, security review, benefit and pricing data feeds.

12. Product Intelligence

- **Usage analytics:** Time per stage, citation clicks, clarification and escalation rates, and where reviewers disagree with the AI. On the patient side: portal views, estimate acceptance, payment choice, declines by reason, callbacks, and appeals.
- **Feedback loop:** Reviewer overrides become labeled examples for the golden set; recurring provider omissions inform intake guidance; cost-related declines inform financial counseling; callback topics show which questions the portal still doesn't answer.
- **Personalization:** None in reviewer decisions in v1; they must be uniform across reviewers. Patients choose language and notification channel.
- **Market signal:** Payers face growing regulatory pressure on prior authorization turnaround and transparency, which favors tools that show their evidence to reviewers and their status to members.

13. Design & UX

- **Landing:** NEW A six-step end-to-end journey at the top, so a first-time viewer understands both sides before choosing one, then two equal entry cards: UM nurse view and Patient view.
- **Interaction model:** Guided flow with a persistent case header, side-by-side checklist and source evidence, and a dark and light theme.

- **Patient portal:** NEW Four tabs: My request, Get ready, Costs & payment, Help. Get ready and Costs unlock on approval. Plain-language copy, English and Spanish, read-aloud, and a mobile-first layout. A "See what the patient sees" link connects the nurse's determination to the patient view.
- **Latency handling:** Staged loading states during extraction; the checklist renders as results arrive; the packet viewer shows the cited page before the rest of the document.
- **Error and uncertainty states:** Dedicated Ambiguous, Missing, and Not Met treatments with the reason shown; a full failure state for unreadable packets.
- **Edit and override:** The nurse can disagree with any criterion status; every action opens a confirmation showing the AI summary that will be stored in the audit trail. A patient can reopen a declined estimate.

14. Technical Implementation Framework

The prototype is a React 19 and Vite single-page app, not Next.js. A production build would follow this shape:

- **Frontend:** React with Tailwind CSS v4; pdfjs-dist for in-browser rendering and sentence highlighting, with on-demand page rendering; Mermaid for workflow diagrams; the browser's speech engine for read-aloud; standard .ics files for calendar export.
- **Packet tooling:** A Python (ReportLab) generator writes the packet PDF and its page index; a check script verifies every citation against the PDF.
- **API layer (target):** Server routes for case retrieval, extraction jobs, audit writes, member status, and payment sessions; streaming for extraction progress.
- **Model integration (target):** Structured-output calls with schema validation and the citation check from section 7.
- **State and data:** Client-side runtime state in the prototype, shared by the nurse and patient views; a database with immutable audit records in production.
- **Deployment:** GitHub Pages via Actions for the demo at sankar.work/Pre-Authorization; hosted app with secrets management and PHI controls for production.

15. Feature-Wise Build Plan

Feature	Phase	Owner	Target Date	Dependencies
Scripted end-to-end demo (six cases)	Phase 1 (done)	PM / builder	Oct 2026	None
106-page packet and patient portal NEW	Phase 1b (done)	PM / builder	Oct 2026	None

Live extraction with citation validation	Phase 2	Engineering	Q1 2027	Model access, de-identified packets
Golden-set evals and shadow mode	Phase 3	PM + clinical SME	Q1 2027	Labeled data
Patient Access API (FHIR) status feed NEW	Phase 3	Engineering	Q1 2027	Payer FHIR server
Backend, roles, and durable audit trail	Phase 4	Engineering	Q2 2027	Security review
Member login, notifications, payments, real estimates NEW	Phase 4	Engineering + billing	Q2 2027	Identity proofing, PCI processor, benefit and pricing feeds
Pilot with one UM team	Phase 5	PM + clinical lead	Q2 2027	Phases 2 to 4, compliance approval

Open Questions

1. What is the acceptable false-Met rate, and who signs off on it?
2. Which policy sources can be licensed or embedded, and under what terms?
3. How should conflicting documents be shown: flag only, or propose a resolution?
4. Which review-tier thresholds should be configurable per payer?
5. What does a defensible audit export need to contain for regulators?
6. NEW Who owns the patient's cost estimate (plan, facility, or both), and how accurate must it be before it's shown?
7. NEW Should the patient see a decision at the same moment the provider does, or after the provider has had a chance to call?
8. NEW Which languages beyond Spanish does the pilot population need?

Assumptions Made

- Format: AI Product PRD, because model grounding, hallucination risk, and evals shape the product.
- The prototype is the MVP; all dates after October 2026 are planning targets, not commitments.
- Baselines and targets are hypotheses, since no real-world measurements exist yet.
- Four criteria and the review-tier thresholds are taken from the current implementation.
- The first user is a payer-side UM nurse; provider-side prior authorization submission is out of scope. The patient portal is payer-operated and member-facing.
- All six cases share one synthetic 106-page packet. Cost figures, appointments and contact details are synthetic, and dates are relative to the day the demo is opened.

- Payments, notifications and callbacks are simulated; nothing leaves the browser. Spanish copy is a draft pending native-speaker review.
- The demo uses synthetic data and is not a clinical tool.

Regulatory references: CMS Interoperability and Prior Authorization Final Rule (CMS-0057-F) fact sheet; CMS, Reconsideration by the Medicare Advantage (Part C) Health Plan.